Projection of Private Values in Auctions

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Taste Projection

- People mispredict others' preferences systematically.
- *Taste projection*: the tendency to overestimate the similarity between others' tastes and one's own.

  $\Longrightarrow$ distorted bidding behaviors in auctions

Motivating Example

A and B bidding for an apartment.

- People's values i.i.d with average \$500k.
- A's valuation: \$300k
- B's valuation: \$700k
- A projects his dislike onto B, estimating B's value $<$ \$500k
- B projects her affection onto A, estimating A's value $>$ \$500k

Motivating Example: FPA

First-price sealed-bid auction (FPA): one's optimal bid $<$ value – bid shading.

- B shades her bid too little.
- A shades his bid too little.
  1. A underestimates B's unconditional expected value, yet
  2. overestimates B's value conditional on it being lower than his own.
- Projection makes them underestimate the heterogeneity in values and exaggerate the competitiveness $\Longrightarrow$ overbidding.

Motivating Example: SPA

Second-price sealed-bid auction (SPA): one's optimal bid $=$ value.

- Misperceptions distort behaviors

  $\iff$ incentives are linked to perceived intensity.

- This link is absent in SPA.

In private-value environments, revenue: FPA $>$ SPA.

Motivating Example: Common Values

Inferential errors emerge when a common-value component exists.

- English auction: suppose each bidder also makes an independent estimate of the maintenance cost.
  1. A and B observe C drops out at a very low price.
  2. A overattributes C's low willingness $\rightarrow$ low private taste.
  3. B overattributes C's low willingness $\rightarrow$ pessimistic info. about the cost.
  4. B may lose even with the highest value.
- The inferential errors may harm efficiency.

Main Results

This paper studies how taste projection affects common auction formats.

Specifically, this paper

- finds that under taste projection,
  1. FPAs yield higher expected revenue than SPAs;
  2. The optimal reserve price is lower than the rational benchmark;
  3. With a common value, SPAs and English auctions are less efficient than FPAs.
- provides a new theoretical explanation for well-documented deviations from rational benchmark predictions.

Model

- One good for auction
- $N \geq 2$ risk neutral bidders, $i \in \{1, 2, \cdots, N\}$
- Independent private values (IPVs) $t_i \in \mathcal{T} = [\underline{t}, \bar{t}]$
- CDF $F : \mathcal{T} \to [0, 1]$, $f = F'$ smooth and > 0
- Hazard rate $\dfrac{f}{1 - F}$ monotone
- Each bidder i simultaneously submits a bid b_i.
- Auction price p
- Bidder i's payoff: $t_i - p$ if he wins and 0 otherwise.

Projection of Private Values

- Building on Gagnon-Bartsch (2016), a player misperceives $F(\cdot) \to \hat{F}(\cdot|t)$.
- LOR's (2003) model of "simple projection bias"
- Player i with realized t_i believes j's value is

$$\hat{t}_j(t_i) \equiv \alpha t_i + (1-\alpha)t_j, \quad \alpha \in [0,1].$$

- Degree of projection α:
  1. $\alpha = 0 \to$ rational benchmark
  2. $\alpha = 1 \to$ each player believes others share his own taste.
- α equal across players

Projection of Private Values

- $\hat{T}(t_i) \equiv \alpha t_i + (1-\alpha)T, T \sim F$
- Misperceived support $\hat{\mathcal{T}}(t_i) = [\underline{t}(t_i), \bar{t}(t_i)]$
- Misperceived CDF

$$\hat{F}(t|t_i) = \Pr\left(\hat{\mathcal{T}}(t_i) \leq t\right) = F\left(\frac{t-\alpha t_i}{1-\alpha}\right), \quad t \in [\underline{t}(t_i), \bar{t}(t_i)]$$

- $\hat{F}(t|t_i)$ inherit assumptions on F.

Perceived distributions

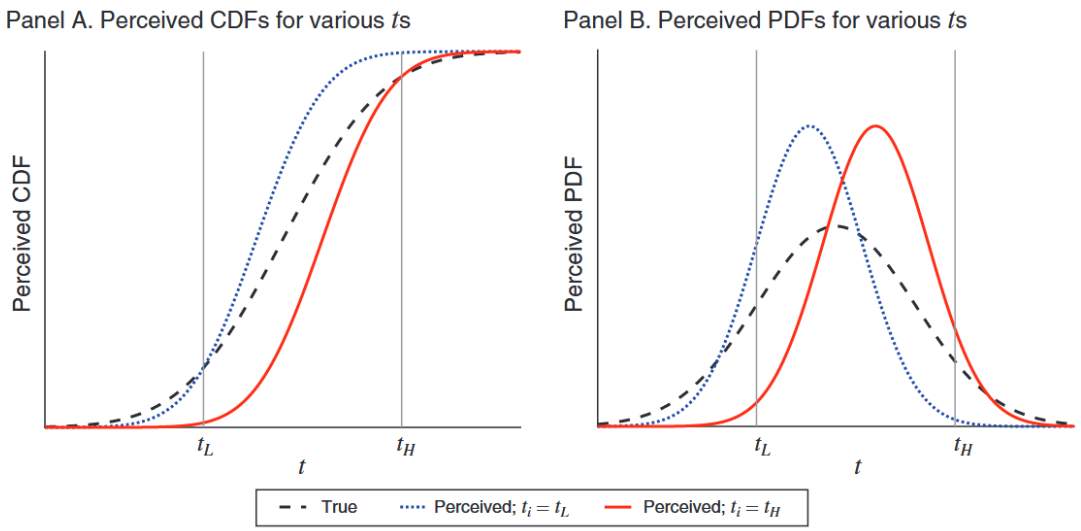
Observation 1

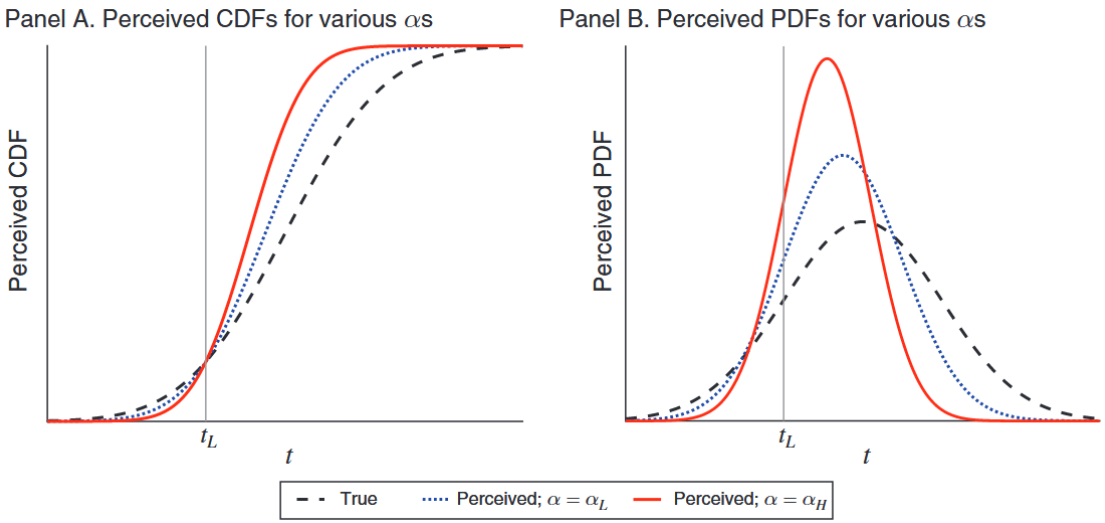
Perceived distributions $\{\hat{F}(\cdot|t)\}_{t\in\mathcal{T}}$ *have the following properties:*

(1) The perceived distribution of a projecting bidder with private value t first-order stochastically dominates that of a bidder with private value $t' < t$.

(2) The perceived distribution of a projecting bidder with private value t is a counterclockwise rotation of the true distribution: $\hat{F}(x|t) < F(x)$ *if* $x < t$; $\hat{F}(x|t) > F(x)$ *if* $x > t$; *and* $\hat{F}(t|t) = F(t)$. *Furthermore, for all* $x \neq t$, $\left|\hat{F}(x|t) - F(x)\right|$ *is strictly increasing in* α.

Perceived distributions





Higher-Order Beliefs

- Naivety: each player neglects that he and others misperceive the distribution.
- A bidder with t believes
  1. All others think private values have CDF $\hat{F}(\cdot|t)$.
  2. This mutual perception is common knowledge.

Solution Concept

- Rational bidders aside from misperceptions about F and naivety
- True auction game Γ
- Perceived auction $\Gamma(\hat{F}(\cdot|t_i))$ with commonly known $\hat{F}(\cdot|t_i)$
- Player i thinks he is in $\Gamma(\hat{F}(\cdot|t_i))$ and that players follow a BNE of it.
- Pure strategy for j in $\Gamma(\hat{F}(\cdot|t_i))$ $\tilde{\beta}^i_j : \hat{T}(t_i) \to \mathbb{R},\ j \in \{1, \cdots, N\}$
- Profile of strategies $\tilde{\beta}^i$
- Set of BNEs of $\Gamma(\hat{F}(\cdot|t_i))$ $B(\Gamma(\hat{F}(\cdot|t_i)))$

Solution Concept

- Player i with t_i predicts that all will bid according to $\hat{\beta}(\cdot|t_i)$, so he bids $b_i = \hat{\beta}(t_i|t_i)$.
- NBE strategy $\hat{\beta}(t) \equiv \hat{\beta}(t|t)$
- *Perceived bidding function* $\hat{\beta}(t|t_i)$ – what he thinks a player with t will bid
- *NBE bidding function* $\hat{\beta}(t)$ – what a player with t actually bids
- Rational benchmark β^*
- Difference from Breitmoser (2019) "type projection equilibrium":
  1. In a TPE, a player believes opponents share his exact type with prob..
  2. Overbidding in a TPE stems entirely from mixing.
  3. Players fully predict and best respond to each other's projection $\leftrightarrow$ naivety assumption

Bidding Stategies: SPA

Proposition 1

In an SPA with IPVs, the strategy profile in which all players bid their private value is an NBE no matter the extent of projection. That is, $\hat{\beta}_{II}(t_i) = t_i$ and $\hat{\beta}$ is thus independent of α.

- Independent of any assumptions on the form of projection

Bidding Stategies: FPA

- In the symmetric rational BNE, Bidder i bids

$$\beta_I^*(t_i) = \mathbb{E}[\max_{j \neq i} T_j \mid \max_{j \neq i} T_j < t_i].$$

Proposition 2

Consider an FPA with IPVs.

(1) The NBE bidding function is $\hat{\beta}_I(t) = \frac{\int_{\underline{t}(t)}^{t} y \hat{f}_1(y|t)\, dy}{\hat{F}_1(t|t)}$.

(2) $\hat{\beta}_I(t) \geq \beta_I^*(t)$*, and this inequality is strict for all* $t > \underline{t}$ *(and* $\alpha > 0$*). Moreover,* $\hat{\beta}_I(t)$ *is increasing in* α*.*

Bidding Stategies: FPA

PROOF For part (1), derive the NBE strategy.

- The highest valuation other than i's: $T_{i,1} \equiv \max_{j \neq i} T_j$
- Player i's perceived CDF and PDF of $T_{i,1}$: $\hat{F}_1(\cdot|t_i)$ and $\hat{f}_1(\cdot|t_i)$, $\hat{F}_1(\cdot|t_i) = \hat{F}(\cdot|t_i)^{N-1}$.
- Perceived symmetric bidding function $\hat{\beta}_I(\cdot|t_i)$: he believes a bidder t solves

$$\max_b \hat{F}_1\left(\hat{\beta}_I^{-1}(b|t_i) \mid t_i\right) \times (t-b),$$

yielding the FOC

$$\frac{\hat{f}_1\left(\hat{\beta}_I^{-1}(b|t_i) \mid t_i\right)}{\hat{\beta}_I'\left(\hat{\beta}_I^{-1}(b|t_i) \mid t_i\right)}(t-b) - \hat{F}_1\left(\hat{\beta}_I^{-1}(b|t_i) \mid t_i\right) = 0. \tag{4}$$

Bidding Stategies: FPA

- The symmetric BNE strategy in $\Gamma(\hat{F}(\cdot|t_i))$: $b = \hat{\beta}_I(t|t_i)$ for all t, so (4) yields:

$$\frac{d}{dt}\left(\hat{F}_1(t|t_i)\hat{\beta}_I(t|t_i)\right) = t\hat{f}_1(t|t_i).$$

- Initial condition $\hat{\beta}_I(\underline{t}(t_i)|t_i) = \underline{t}(t_i)$
- Player i thinks a player with t bid

$$\hat{\beta}_I(t|t_i) = \frac{\int_{\underline{t}(t_i)}^{t} y\hat{f}_1(y|t_i)\,dy}{\hat{F}_1(t|t_i)}$$

- NBE bidding function:

$$\hat{\beta}_I(t_i) = \hat{\beta}_I(t_i|t_i) = \hat{\mathbb{E}}[T_{i,1}|T_{i,1} \leq t_i; t_i] = \frac{\int_{\underline{t}(t_i)}^{t_i} y\hat{f}_1(y|t_i)\,dy}{\hat{F}_1(t_i|t_i)}$$

Bidding Stategies: FPA

For part (2), the NBE bidding function can be rewritten as:

$$\hat{\beta}_I(t_i) = (1-\alpha)\beta_I^*(t_i) + \alpha t_i. \tag{6}$$

- Note that

$$\hat{\beta}_I(t_i) = \frac{\int_{\underline{t}(t_i)}^{t_i} y\hat{f}_1(y|t_i)dy}{\hat{F}_1(t_i|t_i)} = \frac{\frac{1}{1-\alpha}\int_{\underline{t}(t_i)}^{t_i} yf_1\left(\frac{y-\alpha t_i}{1-\alpha}\right)dy}{F_1(t_i)}$$
$$= \frac{\int_{\underline{t}}^{t_i}[(1-\alpha)x + \alpha t_i]f_1(x)dx}{F_1(t_i)}$$

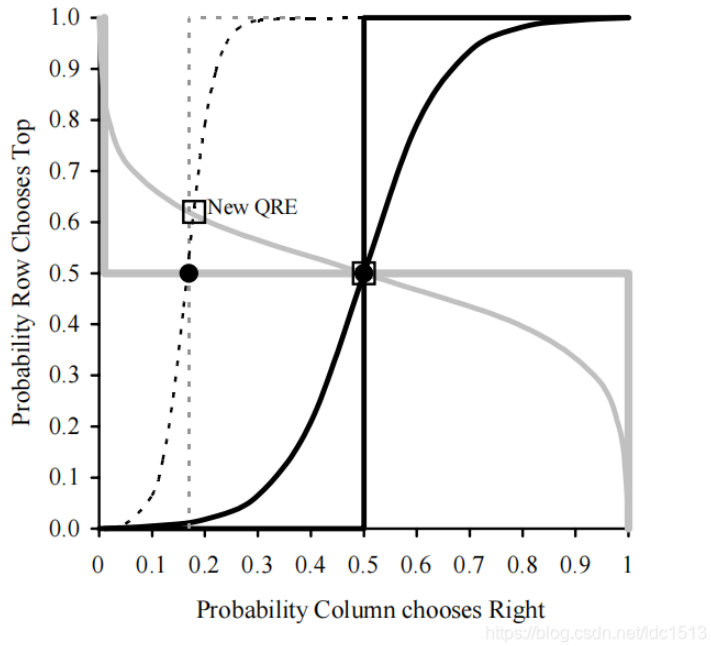
  and $\beta_I^*(t_i) = \dfrac{\int_{\underline{t}}^{t_i} xf_1(x)dx}{F_1(x)}$.

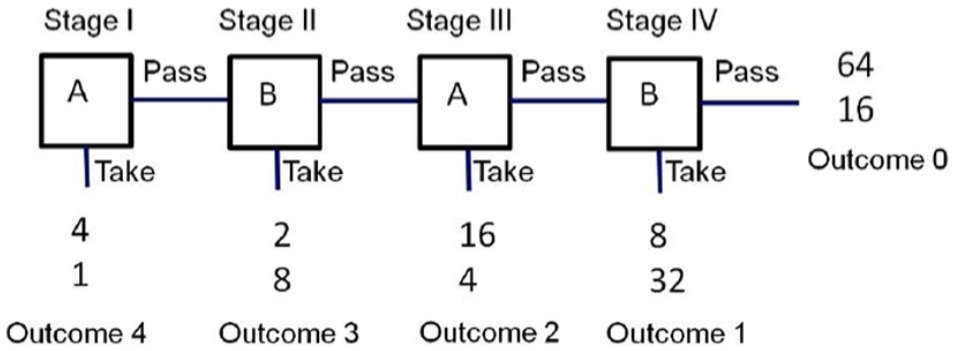
- $\beta_I^*(t_i) < t_i$ when $t_i > \underline{t}$.

Overbidding

- All players overbid $\Rightarrow$ no loss of efficiency
- Difference from other cognitive explanations for overbidding, for example quantal-response equilibrium (Goeree, Holt, and Palfrey 2002) and level-k thinking (Crawford and Iriberri 2007):
  1. Projection explicitly predicts overbidding in FPAs, and no underbidding.
  2. Strategic errors are not a priori but a response to misperceptions.
- Limitations of projection:
  1. A Limited effect in experimental FPAs using induced monetary values.
  2. None can rationalize deviations often observed in SPAs with private values.

QRE and Level-k theory





Revenue Implications

Proposition 3

Consider an auction environment with IPVs.

(1) In an FPA (or Dutch auction), expected revenue is higher with projecting bidders than with rational bidders and is strictly increasing in α.

(2) In an SPA (or English auction), expected revenue with projecting bidders is the same as with rational bidders.

Corollary 1

Revenue equivalence does not hold with projection: if $\alpha > 0$, then an FPA (or Dutch auction) revenue dominates an SPA (or English auction).

Revenue Implications

Proposition 4

Under taste projection, bidders perceive an SPA and an FPA to be payoff equivalent.

- Incorrect beliefs affect expected payments in SPAs.
  1. A projecting bidder in an SPA overestimates the bid of his strongest opponent $\rightarrow$ overestimates the price to be paid.
- Projecting bidders are equally willing to participate in the format preferred by Auctioneer (FPA) and the one that they should prefer (SPA).

Optimal Reserve Price

- Seller (S) knows α and F.
- S's valuation $v_s \in [0, \bar{t})$, independent of t_i
- Rational benchmark $r^* = \dfrac{1 - F(r^*)}{f(r^*)} + v_s$
- Identical in FPA and SPA, independent of N.
- In FPA with projection, the optimal reserve price $\hat{r}(\alpha, N)$ maximizes S's expected revenue:

$$\Pi(r) \equiv N \int_r^{\bar{t}} \hat{\beta}_l(t; r) f(t) F(t)^{N-1} dt + F(r)^N v_s.$$

Rational: $\hat{r}(0, N) = r^*$ for all N.

Optimal Reserve Price

Proposition 5

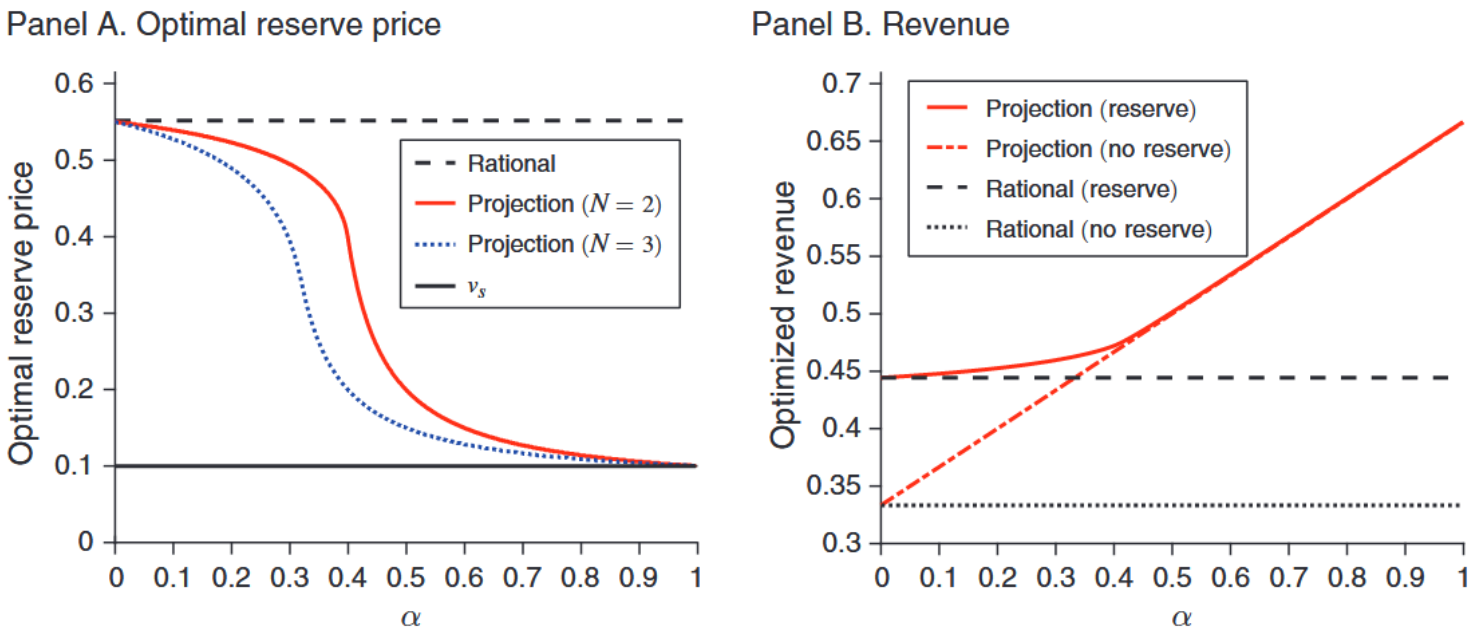
Consider an FPA with IPVs. For all $\alpha > 0$, the revenue-maximizing reserve price $\hat{r}(\alpha, N)$ is strictly lower than r^*. Furthermore, $\hat{r}(\alpha, N)$ is decreasing in both α and N, with $\hat{r}(1, N) = v_s$.

INTUITION

- $\alpha \uparrow$, the effect of r on the strategy of high-value bidders $\downarrow$
  1. Stronger projection weakens the competition-enhancing effect of r on the bids of buyers with $t > r$.
- The prob. of excluding bidders is unaffected.
- S lowers r if $\alpha \uparrow$.

Optimal Reserve Price

- $N \uparrow$, the bidding function of bidders not excluded increases by less than rationally,
- but it has the same effect on the prob. of not selling as rationally.



Optimal Reserve Price

Corollary 2

For all $\alpha > 0$, an FPA with the optimal reserve price is more efficient and yields a higher expected revenue than an SPA with the optimal reserve price. It is also more efficient and yields a higher expected revenue than the optimal auction under rational bidding.

Proposition 6

Consider an FPA with IPVs. There exists an $\tilde{\alpha} \in (0, 1)$ such that, if $\alpha > \tilde{\alpha}$, then the expected revenue under projection with no reserve price is greater than the expected revenue under rational bidding with the optimal reserve price.

- the frequent use of "absolute auction" with no reserve price

Common Values

Model of Goeree and Offerman (2003):

- Bidder i gets the realization of a private signal S_i about the common value.
- Signal i.i.d., with support $\mathcal{S} \equiv [\underline{s}, \bar{s}]$ and CDF $G : \mathcal{S} \to [0, 1]$
- Bidder i' valuation:

$$v_i = t_i + \gamma \sum_{i=1}^{N} s_i,$$

  $\sum_{i=1}^{N} s_i$: the common-value component.
- $\gamma > 0$ commonly known
- Bidders correctly perceive G, projecting tastes but not information.

Common Values

- Aggregate type $\theta_i \equiv t_i + \gamma s_i$
- Bidder i's interim expected valuation:

$$\mathbb{E}[V_i|\theta_i] = \theta_i + \gamma(N-1)\mathbb{E}[S]$$

- True support of aggregate types $\Theta \equiv [\underline{t} + \gamma\underline{s}, \bar{t} + \gamma\bar{s}]$
- Bidder i's perception of Θ:

$$\hat{\Theta}(t_i) \equiv [\underline{t}(t_i) + \gamma\underline{s}, \bar{t}(t_i) + \gamma\bar{s}]$$

- $\mathcal{T}(\theta) \subseteq \mathcal{T}$: set of private values for which θ is plausible.

$$t \in \mathcal{T}(\theta) \Leftrightarrow \theta \in \hat{\Theta}(t)$$

Technical Assumptions

- $f \equiv F'$ and $g \equiv G'$ log-concave
  $\Rightarrow \exists$ a unique symmetric equilibrium where each player's strategy depends only on θ_i.
- Assume $\mu(\theta) \equiv \mathbb{E}[S_i|\theta_i = \theta] - \mathbb{E}[S_i|\theta_i \leq \theta]$ is increasing in θ.
- The relationships extend to projecting bidders' expectations.

Bidding with Common Values

- In SPA,

$$\beta_{II}^*(\theta_i) = \mathbb{E}[V_i|\theta_{i,1} = \theta_i] = \theta_i + \gamma\mathbb{E}[S_j|\theta_j = \theta_i] + \gamma(N-2)\mathbb{E}[S_j|\theta_j \leq \theta_i]$$

$$\hat{\beta}_{II}(\theta_i|t_i) = \theta_i + \gamma\hat{\mathbb{E}}[S_j|\theta_j = \theta_i; t_i] + \gamma(N-2)\hat{\mathbb{E}}[S_j|\theta_j \leq \theta_i; t_i]$$

- In FPA,

$$\begin{aligned}\beta_I^*(\theta_i) &= \mathbb{E}[V_i|\theta_{i,1} \leq \theta_i] - \mathbb{E}[\theta_i - \theta_{i,1}|\theta_{i,1} \leq \theta_i] \\ &= \mathbb{E}[\theta_{i,1}|\theta_{i,1} \leq \theta_i] + \gamma(N-1)\mathbb{E}[S_j|\theta_j \leq \theta_i].\end{aligned}$$

$$\hat{\beta}_I(\theta_i|t_i) = \hat{\mathbb{E}}[\theta_{i,1}|\theta_{i,1} \leq \theta_i; t_i] + \gamma(N-1)\hat{\mathbb{E}}[S_j|\theta_j \leq \theta_i; t_i]$$

Misinference and Competition

Lemma 1

Fixing $\theta \in \Theta$, the following properties hold over the domain $\mathcal{T}(\theta)$:

(1) Misinference Effect: $\hat{\mathbb{E}}[S_j|\theta_j = \theta; t_i]$ and $\hat{\mathbb{E}}[S_j|\theta_j \leq \theta; t_i]$ are decreasing in t_i.

(2) Competition Effect: $\hat{\mathbb{E}}[\theta_{i,1}|\theta_{i,1} \leq \theta; t_i]$ is increasing in t_i.

Efficiency Comparison

Proposition 7

Consider a setting with private and common values.

(1) In an SPA, efficiency with projecting bidders is lower than with rational bidders.

(2) In an FPA, there exists a $\overline{\gamma} > 0$ such that, if $\gamma < \overline{\gamma}$, then efficiency with projecting bidders is higher than with rational bidders.

(3) For all $\alpha > 0$, efficiency is higher in an FPA than in an SPA.

Proposition 8

Consider a setting with private and common values. For any $\alpha > 0$, efficiency in an English auction is (i) lower than with rational bidders, and (ii) lower than in an SPA or FPA.

Efficiency Comparison

- Proposition 7 and 8

  $\Rightarrow$ The English auction is the least efficient, while FPA the best.
- Traditional view: ascending auctions tend to be more efficient than sealed-bid ones.
  1. Following an equilibrium strategy may be easier sequentially rather than simultaneously when players face cognitive limits that impede complex reasoning (e.g., Levin, Kagel, and Richard 1996; Carrillo and Palfrey 2011; Li 2017).

Asymmetric Auctions

Asymmetric Auctions (with private values):

- 2 bidders: a "strong"bidder with valuation drawn from a distribution that first-order stochastically dominates that of a "weak" bidder
  1. SPA/English aution: unchanged
  2. Projection $\Rightarrow$ overbidding in FPAs and increases S's revenue.
  3. Projection increases efficiency in FPAs.

Affiliated Private Values

Affiliated Private Values (APVs):

- With APVs, sufficiently strong projection $\Rightarrow$ revenue: FPA $>$ SPA.
  1. Milgrom and Weber's (1982) "linkage principle": with rational bidders, revenue: SPA $>$ FPA.
- Projection with IPVs $\leftrightarrow$ Rational model with APVs – higher-order beliefs:
  1. A projecting bidder forms incorrect beliefs about others' perceptions of the distribution.
  2. With APVs, a rational bidder is fully aware that those with different private values hold different beliefs about the distribution of values.

More

This presentation omits the following topics:

- Empirical work in auctions and strategic reasoning
- Literature on other psychological biases that are less similar to projection
- Testable model predictions for empirical work

Conclusion and Comment

- This paper studies how taste projection affects bidding, revenue, the optimal reserve price, and efficiency in common auction formats.
- Taste projection sheds light on some stylized deviations in the auction literature and delivers new testable predictions for empirical work.
- It seems to me that incorporating grounded behavioral models into classic theoretic ones is a promising direction.